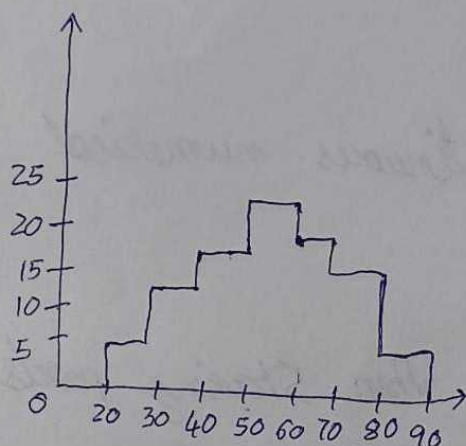
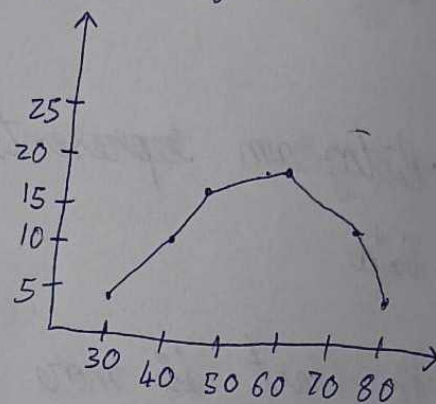


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HOME ASSIGNMENT-1

2400032750

1. CI	20-29	30-39	40-49	50-59	60-69	70-79	80-89
x_i	4	13	18	25	20	14	6

HistogramFrequency Polygon

2. Given Data, 32.2, 32.0, 30.4, 31.0, 31.2, 31.2, 30.3, 29.6, 30.5, 30.7

a) Mean:- $\text{Mean} = \frac{\sum x_i}{n} = \frac{308.9}{10} = 30.89 \text{ mm}$

b) Medium:- Ascending order: 29.6, 30.3, 30.4, 30.5, 30.7, 31.0, 31.2, 32.0, 32.2

Medium = Average of 5th and 6th terms

$$= \frac{30.7 + 31.0}{2} = 30.85 \text{ mm}$$

c) Mode:- The value 31.2 occurs twice
 Mode = 31.2 mm

3. $n = 12$

(i) Range = Max - Min = $0.251 - 0.239$
 $= 0.012$

(ii) Mean = $\bar{x} = \frac{\sum x}{n} = \frac{2.929}{12} = 0.2441$

(ii) Mean Deviation = $\sigma = \sqrt{0.0000127} = 0.00356$

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(ii) Standard Deviation :- $\sigma = \sqrt{0.000127} = 0.00356$

(v) Variance : $\sigma^2 = \frac{\sum (x - \bar{x})^2}{n}$
 $= \frac{0.000152}{12} = 0.0000127$

(vi) C.V = $\frac{0.00356}{0.2441} \times 100$
 $= 1.46$

(vii) Mean = 0.280
S.D = 0.05

CV = $\frac{0.05}{0.280} \times 100$
 $= 17.86$

4. 12, 30, 30, 27, 30, 39, 18, 27, 48, 24, 28

ascending order : 12, 18, 18, 24, 27, 27, 30, 30, 30, 39, 48

(i) Range = 48 - 12 = 36

(ii) Mean = $\frac{303}{11} = 27.65$

(iii) Mean Deviation : $\frac{\sum (x - \bar{x})}{n} = 7.14$

(iv) $Q_1 = 18, Q_3 = 30$

Quartile Deviation = $\frac{30 - 18}{2} = 6$

(v) Variance = $\frac{\sum (x - \bar{x})^2}{n} = 91.34$

(vi) Standard Deviation = $\sqrt{91.34} = 9.56$

(vii) Coefficient of Variance = $\frac{\sigma}{\bar{x}} \times 100$
 $= \frac{9.56}{27.55} \times 100 = 34.70$

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5. (i) Firm A = $986 \times 52.5 = 51,765 \text{ Rs}$

(ii) Firm B = $548 \times 47.5 = 26,030 \text{ Rs}$

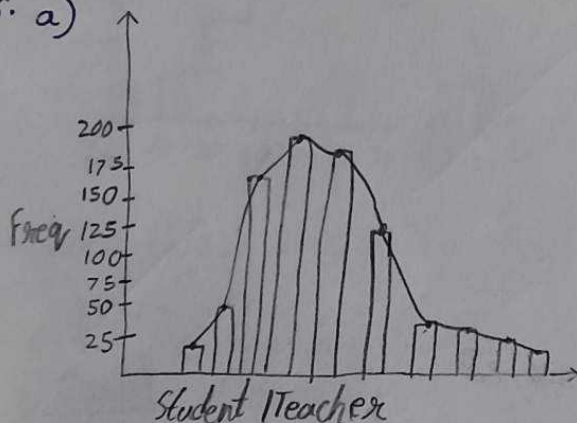
(ii) Variability (CV): -

Firm A: $CV = (10/52.5) \times 100 = 19.05\%$

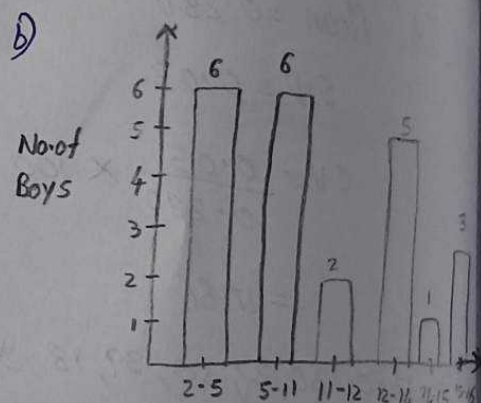
Firm B: $CV = (11/47.5) \times 100 = 23.16\%$

Firm B has greater Usage Variability

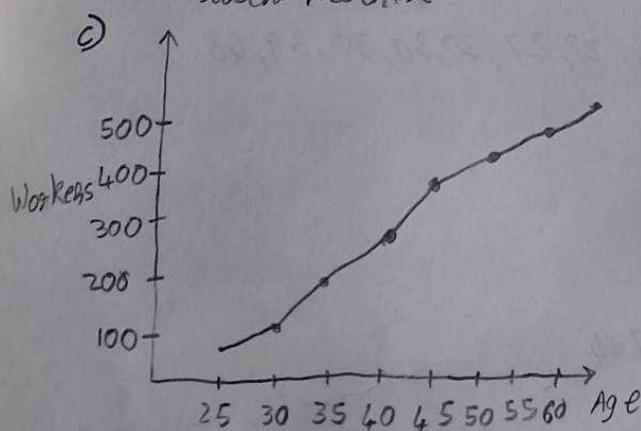
6. a)



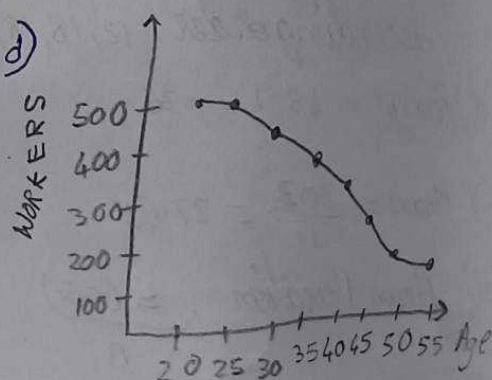
b)



c)



d)



7.

Weight	93-97	98-102	103-107	108-112	113-117	118-122	123-127
No. of Student	2	5	12	17	14	6	3

(i)

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i}$$

$$= \frac{6685}{60} = 111.42 \text{ kg}$$

(ii) Median class = 108-112

Lower Boundary = 107.5

Cumm Freq Before M-Class = 19

Freq of median class = 17

$$\text{Median} = l + \frac{\left(\frac{n}{2} - c\right)}{f} \times h = 107.5 + \frac{30 - 19}{17} \times 5$$

$$= 107.5 + 3.24 = 110.74 \text{ kg}$$

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$$(iii) \text{ Mode} : L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

$$= 107.5 + \frac{5}{8} \times 5$$

$$= 110.63 \text{ kg}$$

8 (i) Mean $A = 55$

$$d = \frac{x - A}{n}$$

$$\sum fd = 84$$

$$\bar{x} = A + h \left(\frac{\sum fd}{N} \right) = 56.55 \text{ Years}$$

(ii) Standard Deviation $\sigma = h \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N} \right)^2}$
 $= 11.96 \text{ Years}$

Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
f_i	5	8	12	15	20	14	12	6

$$N = 92, Q_1 = 20 + \frac{23-13}{12} \times 10 = 28.33$$

a) Median Class = 40-50, $L = 40$, $C = 40$, $f = 20$

$$Q_2 = 40 + \frac{46-40}{20} \times 10 = 43$$

$$Q_3 = 50 + \frac{69-60}{14} \times 10 = 56.43$$

Minimum = 0

$Q_1 = 28.33$

Median = 43

$Q_3 = 56.43$

Maximum = 80

Min = 45

$Q_1 = 75$

Median = 78

$Q_3 = 80$

Max = 82

b) 82, 45, 64, 80, 82

74, 79, 80, 80, 78,

80, 79, 81, 70, 78, 23

$n = 19$

Median (Q_2) = 79

$Q_1 = 73, Q_3 = 80$

10 (ii) Standard Deviation:

$$\sigma_A = \sqrt{\frac{1706}{50} - (5.12)^2}, \sigma_A = 2.81$$

$$\sigma_B = \sqrt{\frac{2146}{50} - (6.16)^2}, \sigma_B = 2.23$$

Coeff. of Variation: $\sim = \frac{\sigma}{x} \times 100$

$$\sigma = \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N} \right)^2}$$

$\sim_A = \frac{2.81}{5.12} \times 100 = 54.88\%$ $\sim_B = \frac{2.23}{6.16} \times 100 = 36.2\%$